Exercise 9.5

(2) (i) continuous for all $x \in \mathbb{R}$ (ii) continuous in $\mathbb{R}$

(iii) continuous for all $x \in \mathbb{R} - (2n+1)\frac{\pi}{2}, n \in z$ (iv) continuous for all $x \in \mathbb{R}$

(v) continuous for $(0, \infty)$ (vi) continuous for all $x \in \mathbb{R} - \{0\}$

(vii) continuous for all $x \in \mathbb{R} - \{-4\}$ (viii) continuous for all $x \in \mathbb{R}$

(ix) continuous for all $x \in \mathbb{R} - \{-1\}$ (x) continuous for all $x \in \mathbb{R} - \frac{n\pi}{2}, n \in z$

(3) (i) not continuous at $x = 3$ (ii) continuous for all $x \in \mathbb{R}$

(iii) continuous for all $x \in \mathbb{R}$ (iv) continuous for all $x \in \left[0, \frac{\pi}{2}\right)$

(4) (i) continuous at $x_0 = 1$ (ii) not continuous at $x_0 = 3$

(6) $\alpha = 4$ (8) 6

(9) (i) not continuous at $x = -1$ and at $x = 1$ (ii) not continuous at $x = 0$

(10) continuous at $x = 0, 1, 3$

(11) (i) removable discontinuity at $x = -2$, $g(x) = \begin{cases} \frac{x^2 - 2x - 8}{x+2} & \text{if } x \neq -2 \\ -6 & \text{if } x = -2 \end{cases}$

(ii) removable discontinuity at $x = -4$, $g(x) = \begin{cases} \frac{x^3 + 64}{x+4} & \text{if } x \neq -4 \\ 48 & \text{if } x = -4 \end{cases}$

(iii) removable discontinuity at $x = 9$, $g(x) = \begin{cases} \frac{3 - \sqrt{x}}{9 - x} & \text{if } x \neq 9 \\ \frac{1}{6} & \text{if } x = 9 \end{cases}$

(12) -2 (13) $f(0) = 0$ (14) $f(1) = \frac{2}{3}$

Exercise 9.6

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
2	3	4	1	1	4	2	2	2	3	4	3	4	3	1
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
1	1	1	4	2	2	2	2	2	4					

Exercise 10.1

(1) (i) 0 (ii) -4 (iii) $-2x$

(2) (i) $f'(1^-) = -1$, $f'(1^+) = 1$, not differentiable (ii) $f'(x) \to -\infty$ as $x \to 1^-$, not differentiable

(iii) $f'(1^-) = 1$, $f'(1^+) = 2$, not differentiable

(3) (i) differentiable (ii) not differentiable

(iii) not differentiable (iv) not differentiable

(5) at $x = -1$ and $x = 8$ are cusps

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